Which Parts of BM25 Help Learned Top-k Selection? Tests in Sparse Retrieval, KV-Cache Page Selection and Expert Routing

Team ANLP Doomsday

Aviral Gupta Mohit Kumar Singh Anirudh Sankar Arihant Tripathy

{aviral.gupta, mohit.singh, anirudh.sankar, arihant.tripathy}@research.iiit.ac.in

IIIT Hyderabad, India https://github.com/avi1o1/ANLP-Doomsday

Abstract

Learned sparse retrieval, KV-cache page selection and mixture-of-experts routing keep the top k items by a learned score. Lexical retrieval corrects its score with BM25's inverse document frequency (IDF), saturation and length normalisation, and prior work attributes BM25's gain on learned features to a Zipf-like feature-frequency curve. We evaluate fourteen sparse codes derived from one frozen dense encoder under all eight on/off combinations of the three components on thirteen collections. The full correction beats the plain inner product in 162 of the 182 code–collection pairs, but a predictor built from the frequency curve and fixed before the experiment does no better than always predicting a gain. Exact Shapley values assign 62% of the gain to saturation, which uses only the document's own values, and tests fixed in advance confirm that its effect follows the spread of the stored values. A rule that uses agreement with text BM25 to decide, without relevance labels, whether IDF and length normalisation help gets the sign right no more often than a constant guess on rows it had not seen and on SPLADE-v3. In KV selection and routing we take an item's frequency to be how often the selector chose it during calibration. With IDF on, the selected KV pages hold 0.21 to 0.44 of full attention's softmax mass, against 0.90 to 0.95 without. Mean RULER accuracy falls from 0.71 and 0.78 (two token budgets) to at most 0.046 on Qwen2.5-1.5B and, in a replication registered before it ran, from 0.43 and 0.46 to at most 0.002 on Llama-3.2-1B. Mean MMLU accuracy on OLMoE falls from 0.480 to at most 0.390. The frequency-curve predictor is wrong in sign on every KV row with a nonzero margin and on 52 of 55 MMLU subjects.

1 Introduction

In lexical retrieval, BM25 adjusts the raw term-count score in three ways. Inverse document frequency (IDF) discounts features that occur throughout the collection, saturation stops rewarding repeated evidence past a point, and length normalisation adjusts for document length (Spärck Jones, 1972; Robertson and Zaragoza, 2009). Three learned selectors score without these corrections. A learned sparse retriever scores a query against a document by an inner product over non-negative features and returns the top k (Formal et al., 2021b; Nguyen et al., 2023). Quest, a KV-cache sparsity method, scores fixed-size pages of the key-value cache against the current query and attends only to the highest-scoring pages (Tang et al., 2024). A mixture-of-experts router scores every expert and sends the token to the top k (Shazeer et al., 2017; Fedus et al., 2022). Each score is non-negative or can be made so, and BM25's correction can be applied to all three (Figure 1). In retrieval the score is a sum over many features. For a KV page or an expert it is a single number, and saturation alone then cannot change the selection (Proposition 1).

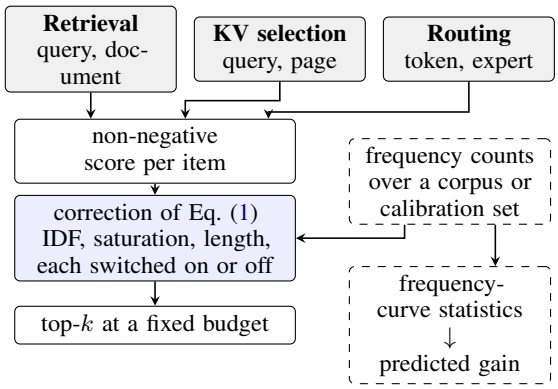


Figure 1: The three settings. Each scores items against an input and keeps the top k. The score is a sum over features in retrieval and one value per page or expert. The BM25 correction is applied between scoring and top-k selection. Its frequency counts are also the input to the frequency-curve predictor of its gain.

Each part of BM25 was designed for words. Applied to learned features, BM25 has both helped and hurt. On one benchmark suite BM25

beats the inner product by 0.20 nDCG@10 on sparse-autoencoder features and loses by 0.11 on SPLADE-v3 (Clavié et al., 2026). The explanation offered is that BM25 helps when the features' document frequencies follow a Zipf-like curve, as word frequencies do (Clavié et al., 2026; Han et al., 2026). If that is right, statistics of the frequency curve should predict the sign of the gain before the experiment is run.

We tested this prediction in retrieval and in two other top-k selections, and it failed in all three. Each test was fixed in a dated commit before its predictions were computed, and we call such tests registered (Table 1). We derive fourteen sparse representations from one frozen dense encoder, multilingual E5-base, and score each under all eight combinations of the three components on thirteen collections. The correction helped almost everywhere, but the registered frequency-curve predictor did not predict the size of the gain. Most of the gain comes from saturation, a component that uses no collection statistics, and the spread of the stored values predicts the size of its effect on held-out parameter settings. Whether IDF and length normalisation help is harder to predict, and a registered rule that decides this without relevance labels, from agreement with text BM25, fails on SPLADE-v3. In KV selection and routing we take an item's frequency to be how often the selector chose it during calibration. IDF then favours items the selector rarely chose, and accuracy falls in both.

Our contributions are:

1. A grid of fourteen representations under all eight configurations on thirteen collections, 1,358 rows, with an exact Shapley decomposition of the gain (Section 5.3).
2. Six registered tests (Table 1). The saturation hypotheses pass, the frequency-curve predictor fails wherever it has a pass mark, and the label-free rule fails on SPLADE-v3 with its registered reference, and fails in KV selection only because a constant guess also scores perfectly there.
3. A mechanism for saturation's gain, confirmed by tests fixed in advance, and evidence that index statistics predict it but not the gain from IDF and length normalisation (Sections 5.4 to 6).
4. The correction in KV selection, on two models, and in routing, with a label-free check by exact attention mass and a proof that, without IDF, saturation cannot change a single-score selection (Sections 3 and 7).

2 Related Work

Term weighting and learned sparse retrieval. IDF began as a statistical account of term specificity (Spärck Jones, 1972). Saturation approximates the two-Poisson model of within-document frequency (Robertson and Walker, 1994), and pivoted length normalisation keeps long documents from being favoured or penalised (Singhal et al., 1996). BM25 combines the three (Robertson and Zaragoza, 2009). Learned sparse retrievers learn term weights and score by inner product, through contextual exact match (Gao et al., 2021) or expansion over the subword vocabulary in SPLADE (Formal et al., 2021b,a; Lassance et al., 2024).

BM25 on learned features. Sparse autoencoders (SAEs) decompose dense activations into many non-negative features (Huben et al., 2024; Gao et al., 2025). Clavié et al. (2026) score SAE features of a frozen dense retriever with BM25 and report 0.582 nDCG@10 against 0.381 for the inner product, and the reverse on SPLADE-v3, 0.455 against 0.568. CL-SR loses 11 to 18 percent of its nDCG@10 on TREC-DL without IDF (Park et al., 2025). Han et al. (2026) find BM25 and the inner product about equal on image features and credit top-k filtering for suppressing what IDF would down-weight. None of these varies the representation systematically or separates all three components.

Predicting effectiveness without labels. Query performance prediction estimates retrieval quality from the query and the ranking (Carmel and Yom-Tov, 2010; Cronen-Townsend et al., 2002; Shtok et al., 2012). Systems can be ranked without judgements against pseudo-relevant documents from pooled output (Soboroff et al., 2001). We apply the same idea to a new target, the difference between two scorers.

Selection inside the transformer. Quest scores KV pages against page min-max key summaries (Tang et al., 2024). H2O keeps recent tokens and heavy hitters, the tokens that have received the most attention (Zhang et al., 2023b). SnapKV keeps the positions recent queries attend to (Li et al., 2024), StreamingLLM the first tokens and a recent window (Xiao et al., 2024), and Expected Attention the positions future queries are predicted to attend to (Devoto et al., 2025). Mixture-of-experts layers route each token to its top-k experts (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022),

balanced by an auxiliary loss (Zoph et al., 2022) or a load-driven bias (Wang et al., 2024a), whose analogue we run as a baseline. We know of no work that applies BM25's components to KV selection or routing.

3 Problem and Registered Tests

The correction. For query and document vectors x_q, x_d with non-negative coordinates we score

$$S(q,d) = \sum_j x_{qj}\, w_j\, T(x_{dj}, \ell_d). \qquad (1)$$

The weight w_j is one or $\mathrm{idf}_j = \ln(1 + (N - df_j + 0.5)/(df_j + 0.5))$, where feature j is nonzero in df_j of the N documents. The length factor ℓ_d is one or $1 - b + bL_d/\overline{L}$, with $L_d = \sum_j x_{dj}$, $\overline{L}$ its mean over the collection, and BM25 parameters k_1 and b. The transform $T(f, \ell)$ is f/ℓ or the saturated $(k_1+1)f/(f+k_1\ell)$. Length normalisation without saturation divides each value by ℓ_d, a variant BM25 does not contain. Each component is on (1) or off (0), and a configuration is written as three bits in the order IDF, saturation, length, so `010` is saturation alone. Configuration `000` (all-off) is the plain inner product and `111` (all-on) is BM25. IDF and length normalisation need collection statistics (document frequencies and mean length), and saturation uses only the document's own values.

Rows and targets. One row pairs a representation with a collection at one support budget (a cap on nonzero features), one pair (k_1, b) and one basis seed, and carries eight nDCG@10 values Q_c, one per configuration c. The 182 primary rows are the fourteen representations on the thirteen collections at the default budget, parameters and seed. The registered target is the margin $m = (Q_{111} - Q_{000})/\,\mathrm{sd}(Q_{000}, \ldots, Q_{111})$, the gain of BM25 over the inner product in units of the spread of the eight scores. It is undefined in three primary rows where all eight scores are equal. The Shapley value of component s averages its marginal effect over every order in which the components can be turned on (Shapley, 1953). Writing Q_A for the configuration with the components in A on,

$$\phi_s = \sum_{A \not\ni s} \frac{|A|!\,(2-|A|)!}{6} \big(Q_{A\cup\{s\}} - Q_A\big), \qquad (2)$$

and the three values sum to $Q_{111} - Q_{000}$. The collection gain $G = Q_{111} - Q_{010}$ is the value of IDF and length normalisation once saturation is applied, and its sign answers the practical question of whether to use collection statistics.

Predictors. The frequency-curve predictor, planned in the proposal and fixed in code on 19 September before any retrieval result, is a ridge regression of m on three diagnostics of the feature-frequency curve (the Zipf slope, the variance of IDF and the activation share of the most frequent 1% of features). The label-free rule replaces Q_c by a pseudo-quality $\tilde{Q}_c$, the nDCG@10 of configuration c's top 100 when the top 10 of a reference ranker on the same query are taken as relevant. The registered reference is text BM25, and the dense encoder and the two fused by reciprocal rank are secondary. The substitution gives estimates $\tilde{\phi}_s$ and $\tilde{G}$ with no fitted parameter, and the rule uses `111` where $\tilde{G} > 0$ and `010` otherwise. A second, fitted predictor regresses ϕ_{idf}, ϕ_{sat}, ϕ_{len} and G on three nested tiers of features. The first holds the pre-registered diagnostics. The second adds index statistics, which describe the nonzero values stored in the index, chiefly their coefficient of variation cv_x. The third, the pseudo-label tier, adds the pseudo-qualities and features of each configuration's top-100 ranking.

Transfer. Eq. (1) applies to any selection with a non-negative score and a frequency count. Each KV page i in a head is one item with one stored value f_i, so the sum in Eq. (1) has one term. The value is Quest's criticality, an upper bound on the query–key product computed from the page's key minima and maxima, passed through a softplus and divided by the head's median selected value during calibration. Calibration runs Quest at 64 positions of the example's own prompt. The page's document frequency is the number of these positions, among those at or after the page, at which Quest selects it, and N is the number of calibration positions at or after the page. The page's length is the norm of its concatenated key minima and maxima. For expert e the value is the router probability, the frequency is the share of calibration tokens routed to e, and the length is the norm of the token's router logits, which all experts of a token share. The selected experts keep their original gate weights. A row is one RULER task at one token budget in KV selection, and WikiText or one MMLU subject in routing. The frequency-curve predictor fitted in retrieval is applied without refitting.

Proposition 1. *When each item carries a single score $f_i \geq 0$, as a KV page or an expert does, and IDF is off, the saturated score is a strictly increasing function of f_i/ℓ_i and selects the same top k as the unsaturated score, up to ties.*

It holds because $(k_1+1)f/(f+k_1\ell) = g(f/\ell)$ with $g(f) = (k_1+1)f/(f+k_1)$, and g is strictly increasing on $f \geq 0$. With IDF on, $\mathrm{idf}_i\, g(f_i)$ and $\mathrm{idf}_i f_i$ can order items differently. In routing all experts of a token share one length, so dividing by it does not change their order either. So the four configurations with IDF off route identically up to ties, as do `100` and `101`.

Registration. The tests in Table 1 come from the code fixed on 19 September (which implements the predictor planned in the project proposal), a hypotheses file (29 September), a registration file (20 October) and an addendum (23 October). The addendum, written after the Qwen results were read and before any Llama run, registered the Llama replication. Each was committed before the predictions it tests were computed. The limit of 1.81 on mean absolute error (MAE) is 1.5 times the frequency-curve predictor's RMSE on m in retrieval with corpora held out (1.21). Always predicting a gain is already right on 90% of the 179 retrieval rows, so our interim report (2 October) added, before any transfer run, the condition of beating that majority sign. The better constant is whichever of always predicting a gain and always predicting a loss is right more often on the rows tested, and the label-free rule must exceed it. Always-on applies `111` in every row, *never* applies saturation alone (`010`) in every row, and the oracle picks the better of the two in each row.

4 Experimental Setup

Representations. All representations come from `multilingual-e5-base` (E5, Wang et al., 2024b) at a pinned revision. Seven bases map an E5 state to a non-negative sparse vector. All but the identity are fitted on the same 200,000 MS MARCO passages (Bajaj et al., 2016). They are the identity, a Gaussian projection to 768 dimensions and PCA to 384, each split into positive and negative parts; soft k-means over 4,096 centroids; product and residual quantisation (8×256, Jégou et al., 2011; Chen et al., 2010); and a Top-K SAE with 8,192 features of which at most 32 are active. They form four basis families for held-out validation (linear, clustered, quantised and learned). Pooled granularity applies a basis to the mean of the token states, and token granularity applies it to each state before summing, so seven bases at two granularities give fourteen representations. In the primary budget the support, the number of nonzero features, is capped at 32 per query and 128 per document, and document frequencies, lengths and IDF come from the capped vectors of each corpus. Retrieval is exact over the whole corpus with $k_1 = 1.2$ and $b = 0.75$.

Collections and sweeps. The thirteen collections come from eleven corpora: MS MARCO dev with TREC-DL 2019 and 2020 on the same documents, six BEIR sets (Thakur et al., 2021), LIMIT (Weller et al., 2026) and the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a). Beyond the primary rows, sweeps vary one setting each (Table 7 in the appendix). SPLADE-v3 (Lassance et al., 2024), with its own backbone, is scored under the same eight configurations as an external comparison.

Significance. All-on is compared with all-off by a paired bootstrap over queries or examples (10,000 draws; 2,000 for SPLADE-v3 and the transfer settings) that recomputes both parts of m, with Holm's correction for multiple comparisons within each pre-declared group of rows (Holm, 1979).

KV-page selection. The attention model is Qwen2.5-1.5B-Instruct (Qwen Team, 2024) in float16, chosen before any run, and Llama-3.2-1B-Instruct is run with the same protocol as a replication registered before it started. All thirteen tasks of the RULER long-context benchmark (Hsieh et al., 2024), from needle retrieval to QA, are run at a 4K context with 100 examples per task and token budgets of 512 and 1,024 over sixteen-token pages. Prefill is dense and decoding attends only to the selected pages. Full attention, Quest and H2O (Zhang et al., 2023b) are run beside the eight configurations, and `000` reproduces Quest's output on every example. The label-free reference is exact attention mass, the share of full-attention softmax mass that falls on the pages a configuration would select, measured along the full-attention generation.

Expert routing. OLMoE-1B-7B (Muennighoff et al., 2025) routes each token to 8 of 64 experts. Quality is the mean token log-likelihood on WikiText-2 (Merity et al., 2017) in 2,048-token windows, reported as perplexity, and zero-shot accuracy on each of the 57 MMLU subjects

Test (registered)	Data	Pass mark	Outcome
Frequency-curve predictor of m (19 Sep.)	179 retrieval rows	none; held-out R^2 reported	R^2 .04, .02 (basis families, corpora held out); sign .90, equal to always predicting a gain
Saturation mechanism, H1–H6 (29 Sep.)	k_1, b sweep	a threshold per hypothesis	all pass (Table 6)
Label-free rule, SPLADE-v3 (20 Oct.)	10 collections	sign above the better constant; rule $\geq$ always-on	fails: sign .50 with BM25; the dense reference meets both (.70)
Frequency-curve predictor, KV (Qwen 20 Oct., Llama 23 Oct.)	24 and 23 RULER rows	sign $\geq$ .70, MAE $\leq$ 1.81, beat the majority sign	fails: sign .00 on both; MAE 4.18, 4.00
Label-free rule, KV (same dates)	24 and 22 RULER rows	as for SPLADE-v3	fails: sign 1.00 equals the better constant; rule equals the oracle
Frequency-curve predictor, routing (20 Oct.)	WikiText, 57 MMLU subjects	as in KV	fails: sign .05, MAE 3.98

Table 1: Tests fixed in a dated commit before their predictions were computed. For SPLADE-v3 the outcomes were already known. Sign is the fraction of rows with a nonzero target whose sign is predicted correctly. The label-free rule of our interim report, evaluated on MS MARCO rows produced afterwards, is reported in Section 5.5 but was not registered.

(Hendrycks et al., 2021), scored by the likelihood of each choice. Frequencies come from 256 training-split windows for WikiText and from each subject's own prompts for MMLU. A load-driven frequency bias, the analogue of loss-free balancing (Wang et al., 2024a), is run beside the eight configurations.

5 Results in Retrieval

5.1 The correction helps the E5 codes

Under the inner product the mean nDCG@10 of the sparse codes, capped at 32 and 128 features, is at most 0.135 on every collection (Table 2). The full correction raises their mean on every collection. Per row, all-on beats all-off in 162 of the 179 primary rows where m is defined (90%), and significantly after Holm correction in 134 of all 182. Under all-on the best code (the token-level SAE on twelve of the thirteen collections) beats text BM25 on six. Yet all-on is the strict best of the eight configurations in only 42 of the 182 rows, and the mean of `011` (saturation and length without IDF, 0.163) is above that of `111` (0.158). The inner product scores below 0.01 in 81 rows, so nearly any correction beats it.

5.2 The frequency-curve predictor predicts almost nothing

The frequency-curve predictor explains almost none of the held-out variance of m, with R^2 of 0.04 when a basis family is held out and 0.02 when a corpus is held out, and its sign accuracy equals that of always predicting a gain. Part of the failure lies in the target. Of the margins, 85% fall between 1.5 and 3.5, because in many rows Q_{000} is near zero and every other configuration is far above it, so m reflects that pattern more than the size of the gain.

5.3 Where the gain comes from

Saturation has the largest Shapley value at every support budget (Table 3). Over the 182 primary rows it takes 62% of the gain, length normalisation 28% and IDF 10%, so collection statistics account for 38% (95% interval 36 to 41%, resampling collections). IDF, the component the Zipfian account is about, contributes least. As supports grow, the share of IDF shrinks and that of length grows, and with uncapped (native) supports IDF's mean Shapley value is slightly negative. Han et al. (2026) report the same trend for image features, where document frequencies converge as more features are active. IDF's small gain does depend on the real frequencies. On ArguAna, NFCorpus and SciFact, shuffling them turns IDF's mean Shapley value negative, and the real value is larger in 83% of the 42 rows (one-sided Wilcoxon $p < 10^{-6}$). Removing IDF altogether, by making all frequencies equal, does not lower all-on nDCG@10 there (0.157 without against 0.147 with, $p = 0.15$).

5.4 What the index predicts

The spread of the stored values (cv_x) predicts saturation's Shapley value. With length normalisation off, saturation maps a stored value f to $g(f)$, and $g(f)/f$ falls as f grows. If every stored value in the index were equal, g would rescale every score by one factor and the ranking would not move. Pooled PQ and RQ are such codes, and their saturation and

	Baselines		E5 sparse codes			SPLADE-v3	
Collection	BM25	E5	000	111	best	000	111
MS MARCO dev	.222	.425	.013	.130	.318	.470	.382
TREC-DL 19	.456	.704	.036	.255	.604	.722	.680
TREC-DL 20	.484	.697	.035	.251	.556	.751	.674
ArguAna	.411	.444	.021	.130	.373	.489	.563
FiQA	.236	.381	.013	.084	.265	.380	.318
NFCorpus	.307	.326	.064	.115	.286	.364	.356
SCIDOCS	.152	.172	.009	.044	.121	.156	.154
SciFact	.662	.694	.088	.197	.426	.717	.716
TREC-COVID	.582	.695	.135	.246	.514	.731	.583
LIMIT	.939	.015	.001	.004	.033	.447	.667
MIRACL hi	.357	.582	.011	.124	.329	.008	.004
MIRACL bn	.407	.700	.025	.217	.488	.000	.001
MIRACL te	.331	.751	.019	.259	.582	.001	.001

Table 2: nDCG@10 of text BM25 and of E5, the dense encoder the codes come from; the mean over the fourteen codes under 000 and 111; the best single code under 111; and SPLADE-v3 under both. MIRACL is outside SPLADE-v3's vocabulary.

Setting	Rows	IDF	Sat.	Len.	Sum
E5 codes, primary	182	.012	.075	.034	.122
E5 codes, 16/64	182	.010	.045	.018	.074
E5 codes, 64/256	182	.013	.095	.056	.163
E5 codes, native	42	−.002	.076	.066	.140
SPLADE-v3	10	.011	.048	−.073	−.013
KV pages, Qwen	24	−.775	−.003	.002	−.777
KV pages, Llama	23	−.509	.000	.006	−.504
experts, MMLU	57	−.094	−.004	−.001	−.099
experts, WikiText	1	−.050	−.002	.000	−.051

Table 3: Mean Shapley value of each component (Eq. 2) and their sum $Q_{111} - Q_{000}$, in each setting's metric (nDCG@10 in retrieval, accuracy in KV selection and MMLU, log-likelihood per token on WikiText). 16/64 and 64/256 are the query and document support caps of the other budgets, and native is uncapped.

length effects are exactly zero in every row (Table 5 in the appendix). The more unequal the values, the more saturation can reorder a ranking for a given k_1 and set of queries. In our codes that reordering usually helps ($\phi_{\text{sat}} > 0$ in 79% of primary rows). We committed six hypotheses on this mechanism before running them on the k_1 and b sweep of five collections, rows not used when the hypotheses were formed (Table 6 in the appendix). All six pass. H5 (no length effect at $b = 0$) is a code check, and H1 overlaps the primary rows. H2, H3, H4 and H6 test the mechanism. In H4 a ridge model of ϕ_{sat} on cv_x fitted at $k_1 = 1.2$ ranks ϕ_{sat} at $k_1 = 0.6$ and 2.4 with Spearman 0.87 and 0.90.

IDF and length normalisation also move the ranking by an amount the index fixes. Whether those moves help depends on whether long documents were over-rewarded and whether rare features separate relevant documents from the rest. For the strongest codes, token-level SAE and token-level PCA, $G > 0$ on fewer than half the collections. Index statistics predict little of G (held-out R^2 of 0.01 to 0.12, Table 8 in the appendix). We take relevance to be the missing input (Section 5.5).

5.5 Supplying relevance without labels

If what the index lacks is relevance, a proxy for relevance should restore the prediction. Only the pseudo-label tier reaches positive held-out R^2 on every target and held-out split, and for G it reaches 0.44 to 0.59, against at most 0.13 without pseudo-labels (Table 8). Within a row the pseudo-quality orders the eight configurations almost as Q_c does (median Spearman 0.98). The rule was designed on the primary rows. On the 181 whose features are all defined, it raises mean nDCG@10 from 0.159 (always-on) to 0.163. The oracle reaches 0.165, so the rule recovers 59% of the difference.

With the BM25 reference, the rule's sign on rows it had not seen is no better than a constant. The interim report fixed the rule before the 84 rows at the two other support budgets on the three MS MARCO collections were produced. On those rows $\tilde{G}$ and G have Spearman 0.91 with the BM25 reference, but the sign is right on 0.87 of rows, against 0.88 for always predicting a gain. With the dense or fused reference it is right on 0.95, but the dense reference is the encoder the codes come from. Always-on scores 0.202 and the oracle 0.203, so at most 0.001 can be gained. The three collections share one corpus, so these rows are not independent.

6 The Correction on SPLADE-v3

SPLADE-v3 under the inner product averages 0.523 nDCG@10, and the correction lowers it on 8 of 10 collections and lowers the mean to 0.509 (Table 2). All-on is significantly worse on four collections and significantly better on ArguAna and LIMIT (bootstrap $p < 0.05$, uncorrected). The direction matches the reversal of Clavié et al. (2026); the size, 0.013 against their 0.113, does not. Inside BM25, length normalisation costs 0.019 (0.528 for `110` against 0.509 for `111`). Its Shapley value, −0.073 (Table 3), is larger because it also averages over `001` and `101`, which divide by length without saturation and score 0.13 and 0.12 below `000` and `100` respectively. Saturation's Shapley value of 0.048 comes from adding it to those two configurations. Added to `000` or `100`, it changes mean nDCG@10 by less than 0.01.

Here the sign of G splits five to five, so a constant sign scores only 0.50. With the registered BM25 reference the label-free rule fails. $\tilde{G}$ has the right sign on five of ten collections and correlates with G at −0.04. On the three MS MARCO collections, adding IDF and length normalisation raises agreement with BM25 by 0.06 to 0.08 while lowering nDCG@10. Agreement with BM25 is higher for configurations that behave like BM25, the bias the registration named in advance. The secondary dense reference meets both marks. Its sign is right on 7 of 10 collections, and its rule scores 0.519, against 0.509 for always-on and 0.515 for *never*. Over ten collections the 95% interval for the gain over always-on includes zero. The fused reference fails.

7 Transfer to KV Pages and Experts

7.1 KV-page selection

On Qwen, variable tracking scores zero under every configuration and under full attention (see the limitations), so its two rows have no defined margin. With IDF on, mean accuracy on Qwen falls to 0.015 to 0.046, against 0.71 and 0.78 for Quest at the two budgets, and the configurations without IDF stay within 0.02 of Quest (Table 4). After Holm correction all-on is significantly worse than all-off in all 24 remaining rows and better in none, and G is negative in all 24. The mean Shapley value of IDF is −0.775, against −0.003 for saturation and 0.002 for length (Table 3).

The registered replication on Llama-3.2-1B-Instruct shows the same pattern. With IDF on, mean accuracy is at most 0.002, against 0.43 and 0.46 for Quest. After Holm correction all-on is significantly worse than all-off in 18 of the 23 rows with a defined margin and better in none, and IDF's mean Shapley value is −0.509. The three undefined rows, where all eight configurations score zero, are both budgets of the first single-needle task and variable tracking at the larger budget.

With IDF on, the selection moves to pages that hold on average less than half of the attention mass. A page's frequency, as we defined it, is Quest's own selection rate during calibration, so IDF down-weights the pages Quest selected most often, and the outcome is partly fixed by this definition. A page selected at all 64 calibration positions gets the smoothed IDF $\ln(1 + 0.5/64.5) \approx 0.008$ and one never selected gets $\ln 130 \approx 4.9$. On the first single-needle task with Qwen at the 1,024 budget, the page holding the needle ranks between second and tenth of 210 to 247 pages by selection rate, so IDF gives it one of the lowest weights. Exact attention mass shows this on every task. Averaged over tasks, the pages a configuration with IDF on would select hold 0.29 to 0.44 of the softmax mass on Qwen and 0.21 to 0.32 on Llama, against 0.90 to 0.95 without IDF, and on every task each configuration with IDF on holds less mass than any without.

As Proposition 1 predicts, saturation without IDF changes the task score almost nowhere. On Qwen, `010` and `000` differ in task score on 38 of 2,600 example–budget pairs and in output on at least 448. We attribute both to float16 ties.

The registered frequency-curve predictor fails on both models. On Qwen it predicts a gain on all 24 rows and is wrong on every one, with an MAE of 4.18 against the limit of 1.81. On Llama it is wrong on every row with a nonzero margin, with an MAE of 4.00. Its predictions on Qwen range only from 2.13 to 2.21. It predicts a gain for nearly any input, so this failure says little about page frequency.

The registered label-free rule uses exact attention mass as its reference. Since IDF's pages always hold less mass, $\tilde{G} < 0$ on every row where the reference exists (26 on Qwen, 24 on Llama). On the rows with $G \neq 0$ (24 and 22) its sign is always right, and the retrieval majority sign is never right. The configurations it picks reach the oracle's mean accuracy (0.742 over the 26 Qwen rows, 0.484 over the 24 Llama rows with a reference), against 0.024 and 0.001 for always-on. It still fails the registered test. With no row having $G > 0$, always predicting

Model		Budget	Full	Quest	H2O	000	001	010	011	100	101	110	111
Qwen	Accuracy	512	.811	.705	.181	.705	.719	.707	.718	.026	.015	.023	.022
		1024	.811	.778	.266	.778	.771	.777	.771	.036	.046	.026	.027
	Mass	512	1	.904	–	.904	.916	.904	.916	.327	.333	.291	.294
		1024	1	.942	–	.942	.950	.942	.950	.437	.443	.404	.407
Llama	Accuracy	512	.494	.433	.091	.433	.448	.433	.448	.000	.002	.002	.002
		1024	.494	.459	.136	.459	.465	.459	.465	.000	.000	.000	.000
	Mass	512	1	.896	–	.896	.912	.896	.912	.241	.245	.208	.210
		1024	1	.938	–	.938	.947	.938	.947	.310	.316	.272	.275

Table 4: Mean RULER accuracy (13 tasks, 4K context, 100 examples each) and mean share of full attention's softmax mass on the pages each configuration would select, at token budgets of 512 and 1,024. Full is full attention, and Quest equals 000 on every example. Llama's mass excludes the first single-needle task, where full attention gives an empty answer.

a loss also scores 1.00, and the pass mark requires exceeding it. We did not foresee this for Qwen, and the Llama addendum stated it in advance. Across rows, $\tilde{G}$ and G have Spearman -0.32 on Qwen and -0.03 on Llama, so the rule gets the direction but not the size. Output agreement with full attention, the secondary reference, gives the same decisions.

7.2 Expert routing

Configuration 000 is the native forward pass. As Section 3 predicts, 001 to 011 give the same result as 000 on every example, and 101 the same as 100 (Table 9 in the appendix). With IDF on, mean MMLU accuracy falls from 0.480 to between 0.382 and 0.390, and WikiText perplexity rises from 7.49 to 7.86 and 7.88. All-on is worse than all-off on 52 subjects and better on 3, and significantly worse after Holm correction in 22 of the 58 rows (57 subjects and WikiText). As in KV selection, an expert's frequency is the router's own selection rate during calibration, so IDF sends tokens towards experts the router rarely chooses. The frequency bias, which adds a load-balancing term to the router probability instead of multiplying it, loses less (MMLU 0.428, perplexity 7.73). The frequency-curve predictor gives 2.05 to 2.18 for every subject, and its sign is right on 3 of the 55 subjects with a nonzero margin (0.05), with an MAE of 3.98 against the limit of 1.81. It is also wrong in sign on WikiText. No label-free rule was registered for routing, because its planned reference, the output of a mixture over all 64 experts, was not implemented.

8 Discussion

Index statistics predict little of G in our data, and the label-free rule depends on its reference. Agreement with BM25 is higher for configurations that behave like BM25, so on SPLADE-v3 it favours the wrong ones. Exact attention mass picks the right setting on every KV row with a reference, but so does a constant, and $\tilde{G}$ does not track the size of G (Spearman -0.32 on Qwen and -0.03 on Llama).

9 Conclusion

BM25's correction helps the E5 sparse codes mostly through saturation, whose effect follows the spread of the stored values, and hurts SPLADE-v3 mostly through length normalisation. The frequency curve does not predict whether collection statistics help. Agreement with BM25 explains about half of the held-out variance of G on the E5 codes, but as an unfitted rule its sign is no better than a constant on new rows. With frequency taken as the selector's own selection rate, IDF moves KV selection to pages holding on average 0.21 to 0.44 of the attention mass and lowers accuracy in KV selection on two models and in expert routing.

Challenges, Limitations and Future Work

One encoder underlies every retrieval code, E5 saw MIRACL in training, and SPLADE is one model. The KV runs use one 4K context and 100 examples per task, not the planned 500. Variable tracking lacks RULER's answer prefix and scores about zero in every configuration. Llama's full-attention answer is empty on the 100 first single-needle examples and 262 others, mostly QA and multi-key, so its mass reference covers 938 of 1,300. The label-free sign mark cannot be met when no row has $G > 0$. Routing uses one model. A frequency counted over unrelated prompts is the next test.

References

Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. 2016. MS MARCO: A human generated machine reading comprehension dataset. *arXiv preprint arXiv:1611.09268*.

David Carmel and Elad Yom-Tov. 2010. *Estimating the Query Difficulty for Information Retrieval*. Synthesis Lectures on Information Concepts, Retrieval, and Services. Morgan & Claypool.

Yongjian Chen, Tao Guan, and Cheng Wang. 2010. Approximate nearest neighbor search by residual vector quantization. *Sensors*, 10(12):11259–11273.

Benjamin Clavié, Sean Lee, Aamir Shakir, and Makoto P. Kato. 2026. Latent terms: Dense retrievers contain trivially extractable BM25-ready Zipfian vocabularies. *arXiv preprint arXiv:2605.29384*.

Steve Cronen-Townsend, Yun Zhou, and W. Bruce Croft. 2002. Predicting query performance. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 299–306. ACM.

Alessio Devoto, Maximilian Jeblick, and Simon Jégou. 2025. Expected attention: KV cache compression by estimating attention from future queries distribution. *arXiv preprint arXiv:2510.00636*.

William Fedus, Barret Zoph, and Noam Shazeer. 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39.

Thibault Formal, Carlos Lassance, Benjamin Piwowarski, and Stéphane Clinchant. 2021a. SPLADE v2: Sparse lexical and expansion model for information retrieval. *arXiv preprint arXiv:2109.10086*.

Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. 2021b. SPLADE: Sparse lexical and expansion model for first stage ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2288–2292. ACM.

Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2025. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations (ICLR)*.

Luyu Gao, Zhuyun Dai, and Jamie Callan. 2021. COIL: Revisit exact lexical match in information retrieval with contextualized inverted list. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 3030–3042. Association for Computational Linguistics.

Donghoon Han, Eunhwan Park, and Seunghyeon Seo. 2026. Visual words meet BM25: Sparse autoencoder visual word scoring for image retrieval. *arXiv preprint arXiv:2603.05781*.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. In *International Conference on Learning Representations*.

Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, and Boris Ginsburg. 2024. RULER: What's the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*.

Robert Huben, Hoagy Cunningham, Logan Smith, Aidan Ewart, and Lee Sharkey. 2024. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations (ICLR)*.

Hervé Jégou, Matthijs Douze, and Cordelia Schmid. 2011. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1):117–128.

Carlos Lassance, Hervé Déjean, Thibault Formal, and Stéphane Clinchant. 2024. SPLADE-v3: New baselines for SPLADE. *arXiv preprint arXiv:2403.06789*.

Dmitry Lepikhin, HyoukJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. 2021. GShard: Scaling giant models with conditional computation and automatic sharding. In *International Conference on Learning Representations (ICLR)*.

Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. Pointer sentinel mixture models. In *International Conference on Learning Representations*.

Niklas Muennighoff, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Jacob Morrison, Sewon Min, Weijia Shi, Pete Walsh, Oyvind Tafjord, Nathan Lambert, Yuling Gu, Shane Arora, Akshita Bhagia, Dustin Schwenk, David Wadden, Alexander Wettig, Binyuan Hui, Tim Dettmers, Douwe Kiela, and 5 others. 2025. OLMoE: Open mixture-of-experts language models. In *International Conference on Learning Representations (ICLR)*.

Thong Nguyen, Sean MacAvaney, and Andrew Yates. 2023. A unified framework for learned sparse retrieval. In *European Conference on Information Retrieval (ECIR)*, pages 101–116. Springer.

Seongwan Park, Taeklim Kim, and Youngjoong Ko. 2025. Decoding dense embeddings: Sparse autoencoders for interpreting and discretizing dense retrieval. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 26468–26485, Suzhou, China. Association for Computational Linguistics.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.

Stephen E. Robertson and Steve Walker. 1994. Some simple effective approximations to the 2-Poisson model for probabilistic weighted retrieval. In *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '94)*, pages 232–241. Springer.

Lloyd S. Shapley. 1953. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games, Volume II*, number 28 in Annals of Mathematics Studies, pages 307–317. Princeton University Press.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and Jeff Dean. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*.

Anna Shtok, Oren Kurland, David Carmel, Fiana Raiber, and Gad Markovits. 2012. Predicting query performance by query-drift estimation. *ACM Transactions on Information Systems*, 30(2):1–35.

Amit Singhal, Chris Buckley, and Mandar Mitra. 1996. Pivoted document length normalization. In *Proceedings of the 19th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 21–29. ACM.

Ian Soboroff, Charles Nicholas, and Patrick Cahan. 2001. Ranking retrieval systems without relevance judgments. In *Proceedings of the 24th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 66–73. ACM.

Karen Spärck Jones. 1972. A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation*, 28(1):11–21.

Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. 2024. Quest: Query-aware sparsity for efficient long-context LLM inference. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235, pages 47901–47911. PMLR.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks (NeurIPS Datasets and Benchmarks)*.

Lean Wang, Huazuo Gao, Chenggang Zhao, Xu Sun, and Damai Dai. 2024a. Auxiliary-loss-free load balancing strategy for mixture-of-experts. *arXiv preprint arXiv:2408.15664*.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024b. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

Orion Weller, Michael Boratko, Iftekhar Naim, and Jinhyuk Lee. 2026. On the theoretical limitations of embedding-based retrieval. In *International Conference on Learning Representations (ICLR)*.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*.

Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2023a. MIRACL: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics*, 11:1114–1131.

Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023b. H2O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 34661–34710.

Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam Shazeer, and William Fedus. 2022. ST-MoE: Designing stable and transferable sparse expert models. *arXiv preprint arXiv:2202.08906*.

A Further results

Basis	Gran.	111	cv_x	ϕ_{sat}	IDF + len.	$G > 0$
identity	pooled	.148	0.36	.044	.68	1.00
	token	.142	0.57	.061	.55	.77
Gaussian	pooled	.098	0.28	.020	.77	.85
	token	.088	0.51	.030	.64	.92
PCA	pooled	.221	0.71	.138	.24	.69
	token	.310	1.22	.155	.45	.46
k-means	pooled	.018	0.05	.000	–	.31
	token	.185	1.65	.125	.11	.85
PQ	pooled	.012	0.00	.000	1.00	.62
	token	.144	1.36	.099	.09	.69
RQ	pooled	.056	0.00	.000	1.00	.92
	token	.292	1.97	.176	.18	.69
SAE	pooled	.127	0.18	.014	.66	.77
	token	.375	3.02	.193	.32	.46

Table 5: Per representation (Gran. is granularity), over its 13 primary rows, the mean all-on nDCG@10, mean cv_x, mean saturation Shapley value, share of the gain from IDF and length, and fraction of collections with $G > 0$. Pooled k-means has no gain to share.

	Hypothesis	Result
H1	IDF + length share at default in $[.36, .42]$	.393
H2	ϕ_{sat} at $k_1 = 0.6$ exceeds $k_1 = 2.4$	$.093 > .080$
H3	within-corpus $\rho(\mathrm{cv}_x, \phi_{\mathrm{sat}}) \geq 0.3$	.89, .89
H4	cv_x model ranks ϕ_{sat} at $\rho \geq 0.5$	.87, .90
H5	$\phi_{\mathrm{len}} = 0$ exactly at $b = 0$	max 0.0
H6	ϕ_{len} at $b = 0.75$ exceeds $b = 0.5$	$.045 > .029$

Table 6: Confirmatory tests, committed before they were run, on the k_1 and b sweep. ρ is Spearman, and the default is $k_1 = 1.2$, $b = 0.75$. H2 and H6 have Holm-corrected $p = 3.6 \times 10^{-5}$. H3 and H4 are tested at $k_1 = 0.6$ and 2.4; H4's model is fitted at $k_1 = 1.2$. H5 covers 210 rows.

Sweep	Coll.	Rows
primary, 14 representations	13	182
support budgets 16/64, 64/256	13	364
native support	3	42
$k_1 \in \{.6, 1.2, 2.4\}$, $b \in \{0, .5, .75\}$	5	560
basis seeds 1 and 2	5	100
shuffled, uniform *df* controls	3	84
shared SAE dictionary	13	26
all rows	13	1,358

Table 7: Retrieval rows, each with eight configurations. Rows a sweep shares with the primary rows are counted once, so the rows sum to the total. Native support is uncapped, and the shared SAE dictionary applies the token-level SAE dictionary at both granularities.

Target	Features	Family	Corpus	Params
ϕ_{sat}	pre-registered	$-.02$	.16	.06
	index	.33	.29	.56
	pseudo-labels	.66	.37	.77
ϕ_{len}	pre-registered	$-.30$	$-.25$	.01
	index	$-.13$	.25	$-.09$
	pseudo-labels	.59	.68	.82
ϕ_{idf}	pre-registered	$-.07$	$-.05$	.02
	index	$-.05$	.04	.08
	pseudo-labels	.37	.33	.76
G	pre-registered	.01	.07	.13
	index	.01	.03	.12
	pseudo-labels	.53	.44	.59
	pseudo-labels, E5	.63	.56	.78

Table 8: Held-out R^2 of ridge regression on the 181 primary rows whose features are all defined, holding out basis families or corpora, or predicting the 560 rows at other k_1 and b. The index tier adds index statistics to the pre-registered features, and the pseudo-label tier adds ranking features and pseudo-qualities. Pseudo-labels use text BM25 unless marked E5 (the dense reference).

Routing	MMLU acc.	WikiText ppl.
native, 000–011	.480	7.49
frequency bias	.428	7.73
100, 101	.390	7.86
110	.386	7.88
111	.382	7.88

Table 9: OLMoE-1B-7B: mean zero-shot accuracy over the 57 MMLU subjects (14,042 questions) and WikiText-2 perplexity, for the native router, the load-driven frequency bias and the eight configurations. Configurations in one row route identically.